Fabric Scanner: Spatial, Frequency and Flash-Sheen Fusion with Retrieval-Grounded Voice Q&A for Visually Impaired Shoppers

Sneha Sridhar

School of Computer Science and Engineering
VIT University
Registration No. 24BCE2103
sneha.sridhar2024@vitstudent.ac.in

ABSTRACT

Blind and low-vision shoppers cannot use the visual cues, such as sheen, weave and surface texture, that sighted people use to tell fabrics apart, and generic scene-description tools are not designed for fine-grained material recognition. We present Fabric Scanner, a prototype that identifies a fabric from a close-up capture and then answers spoken follow-up questions about it. A three-branch network fuses an ImageNet-pretrained ResNet18 spatial branch, a 2D discrete wavelet transform (DWT) frequency branch, and a sheen branch that operates on the difference between a flash-OFF and a flash-ON image. A hybrid conversational layer routes each question either to a fabric knowledge base conditioned on the classifier output or to the Moondream2 vision-language model, with Whisper for speech input and spoken answers, all behind a browser interface with a hold-steady capture trigger. On six classes of the iBUG Fabrics Dataset (1,132 samples) the model reaches 81.7% accuracy and 0.76 macro-F1 on a held-out test set of 169 samples. A controlled ablation shows that neither the DWT branch nor the sheen branch gives a measurable gain over the spatial branch alone (accuracy 0.78 to 0.82 across variants, within a margin of error of about six points), so we report their benefit as unproven. Accuracy on live camera captures and usability with blind users have not yet been evaluated.

CCS Concepts

• Human-centered computing→Accessibility technologies • Computing methodologies→Computer vision • Computing methodologies→Natural language processing

Keywords

Fabric recognition; assistive technology; visually impaired; wavelet transform; multi-flash imaging; vision-language model; retrieval-grounded question answering

1. INTRODUCTION

Independent access to sensory information is essential for visually impaired people to carry out everyday tasks such as shopping without depending on others. Identifying a fabric is a particular difficulty. Sighted shoppers judge cotton, silk, wool, denim or a synthetic blend largely by sheen, weave density and surface texture, and none of these cues is available to a blind shopper. Existing assistive tools mostly offer generic scene description or object detection, which is not designed for fine-grained texture discrimination, especially when two materials share colour and shape but differ in surface structure and reflectance.

This project addresses that gap with a system that identifies a fabric from a close-up capture and then lets the user ask follow-up questions by voice, such as whether the material suits summer or how to wash it, and hear a spoken answer. The aim is to give the shopper an independent way to check what a fabric is and whether it fits their need, instead of relying on shop staff or a companion.

The design responds to three requirements raised in review of the project abstract: (i) fuse multi-scale spatial and frequency information, (ii) capture specular reflection with a multi-flash protocol, and (iii) ground a vision-language model in the classifier output rather than letting it answer unaided. We implemented all three and tested them, and we report honestly what the measurements do and do not show. This paper makes the following contributions:

(1) A three-branch fusion classifier combining spatial, DWT frequency and flash-difference sheen features for six fabric classes, trained with a class-weighted loss and evaluated on a held-out split.

(2) A hands-free capture protocol: a hold-steady trigger followed by a flash-OFF and flash-ON pair (the phone torch, or a full-screen white flash on a laptop), aligned and differenced to give a sheen map.

(3) A hybrid retrieval-grounded voice question-answering layer, designed after documenting how a small vision-language model fails when used alone, together with a browser-based implementation that integrates every stage.

(4) A controlled branch ablation, reported without overstating it: on this dataset neither the frequency nor the sheen branch gives a measurable improvement, and we discuss why.

2. RELATED WORK

2.1 Frequency-Domain Features for Texture and Material Classification

Texture has long been analysed with multi-resolution tools because the information that separates textures is spread across frequency bands rather than raw pixel values. Arivazhagan and Ganesan showed that statistical and co-occurrence features computed on wavelet sub-bands classify textures effectively [1]. With deep learning, attention turned to the fact that convolution and pooling in a CNN act as a limited form of spectral analysis. Fujieda et al. generalised these layers with wavelet transforms in their wavelet CNN and reported better texture accuracy with fewer parameters [5], and Li et al. integrated wavelet decomposition into standard CNNs to improve robustness to noise [12]. Most closely related, Fortuna-Cervantes et al. proposed Discrete Wavelet Transform Pooling because ordinary pooling discards information in images with repetitive patterns, and

evaluated it for texture and material classification on CIFAR-10, DTD and FMD [4]. These studies motivate our frequency branch, which feeds the high-frequency DWT sub-bands to a dedicated network in parallel with a pretrained spatial CNN.

2.2 Multi-Flash Imaging and Specular Reflectance

Computational photography has long used several differently lit captures of one surface to separate specular from diffuse reflectance. Feris et al. used multi-flash photography to detect and reduce specular highlights, exploiting the fact that highlights move with the light position while diffuse shading does not [3]. We reuse the same physical principle in the opposite direction: the specular component is treated as a discriminative signal rather than an artefact, since sheen is one of the main cues separating fabrics such as cotton, polyester and silk.

2.3 Visual Question Answering for Visually Impaired Users

Assistive question answering began with VizWiz, which routed photos taken by blind users to human workers for near real-time answers [2], and later produced the VizWiz Grand Challenge dataset of visual questions asked by blind people [6]. More recent systems use vision-language models directly. Hao et al. combine an image-tagging model with the InstructBLIP vision-language model through prompt engineering, use Whisper for speech input, and report in an ablation that the vision-language model alone produced details not present in the image while adding the recogniser's tags removed many of them [7]. This supports the principle behind our design: a language model answers more reliably when it is conditioned on the output of a dedicated recogniser. We apply that principle with a fabric classifier and add retrieval of curated fabric facts, in the spirit of retrieval-augmented generation [11], for the questions where a small model is least dependable.

2.4 Gap Addressed

In the sources reviewed, texture-oriented work focuses on classification accuracy, multi-flash work on highlight removal, and assistive question answering on scenes and objects. We did not find a single system that combines frequency-domain texture features, a flash-difference sheen cue and a classifier-grounded conversational layer aimed at fabric choice for blind shoppers. This project is a first prototype of that combination, and the ablation in Section 5 tests whether the added cues help.

3. SYSTEM DESIGN

We set out five design goals for an accessible and dependable fabric assistant:

(1) **D1: Hands-free capture.** The user cannot confirm framing visually, so the system detects a steady hold itself and gives spoken guidance.

(2) **D2: Sheen as an explicit signal.** A flash-OFF and flash-ON pair is differenced so that reflectance is available to the classifier as its own input.

(3) **D3: Complementary features.** Coarse appearance, fine weave structure and reflectance are encoded by separate branches and fused.

(4) **D4: Consistent, grounded answers.** Property questions are answered from curated facts for the predicted fabric, and the vision-language model is used only where it is dependable.

(5) **D5: Low-setup delivery.** The whole interaction runs in a web browser on a laptop or an Android phone.

3.1 Interaction Flow

Figure 1 shows the architecture. The user opens the page and starts the camera. When the picture has been steady for 1.2 seconds the system takes one photo in room light and one under flash, and the server aligns the pair, builds the sheen map and classifies the fabric. The result is spoken, with the confidence, or with the two leading candidates when the top probability is below 60%. The assistant then beeps and listens. Each spoken question is transcribed, routed, answered and spoken back, and the user says “done” to end the conversation and scan another fabric. Typed questions and question chips offer the same function without voice.

3.2 Capture Protocol

In the browser, frames are sampled every 100 ms and reduced to 128×96 grey levels. To make the steadiness test robust to sensor noise, movement is measured on a 32×24 picture obtained by averaging 4×4 blocks: it is the mean absolute difference between consecutive pictures, smoothed with an exponential average (weight 0.4 on the newest value). A scan is triggered once the movement stays below a limit for 1.2 s. The limit is 6 grey levels by default, with strict (3) and relaxed (12) settings for a camera on a stand or held by hand. Brightness and sharpness do not block a scan, because fabric held very close to the lens is naturally dimmer and softer: a dim (mean grey below 30) or soft picture produces a spoken and on-screen tip, and only a covered lens (mean grey below 8) is reported and blocks the scan. If the picture does not settle within 8 s the app captures anyway, and it gives up after 40 s. These values were set by hand and have not been tuned or validated. For the flash, the page uses the phone torch through the browser's torch constraint where the device exposes it, and otherwise turns the whole screen white, waiting 0.6 to 0.75 s for exposure to settle before the second frame. The frames are sent to the server as JPEG (quality 0.92) of a centred square crop of at most 640 pixels per side. The crop can be narrowed with a zoom setting (1x, 1.5x, 2x or 3x; default 2x) to approximate the close-up scale of the training patches. The server computes the mean absolute grey-level change between the two frames, and if it is below 2 it warns that the flash barely changed the picture, because the sheen map then carries little information.

3.3 Capture Hardware

Live capture was run in Google Chrome on a laptop with its built-in HP Wide Vision HD camera (the device name reported by the browser). The page requests 1280×720 video, and the browser reported that resolution as delivered. This camera has no flash or torch, so the screen-flash path was used, with the fabric held in front of the lens and the display as the light source. The phone-torch path, which needs an Android phone and an HTTPS connection, is implemented but has not been exercised. In three live scans of fabric held a few centimetres from the lens, the screen flash changed the mean grey level of the picture by 12.06, 19.24 and 20.5, well above the warning threshold of 2, so the flash produced a measurable difference between the two frames. A scan with the fabric pressed against the lens gave only 2.62 and almost black images, because no light reaches the fabric. No other cameras were tested.

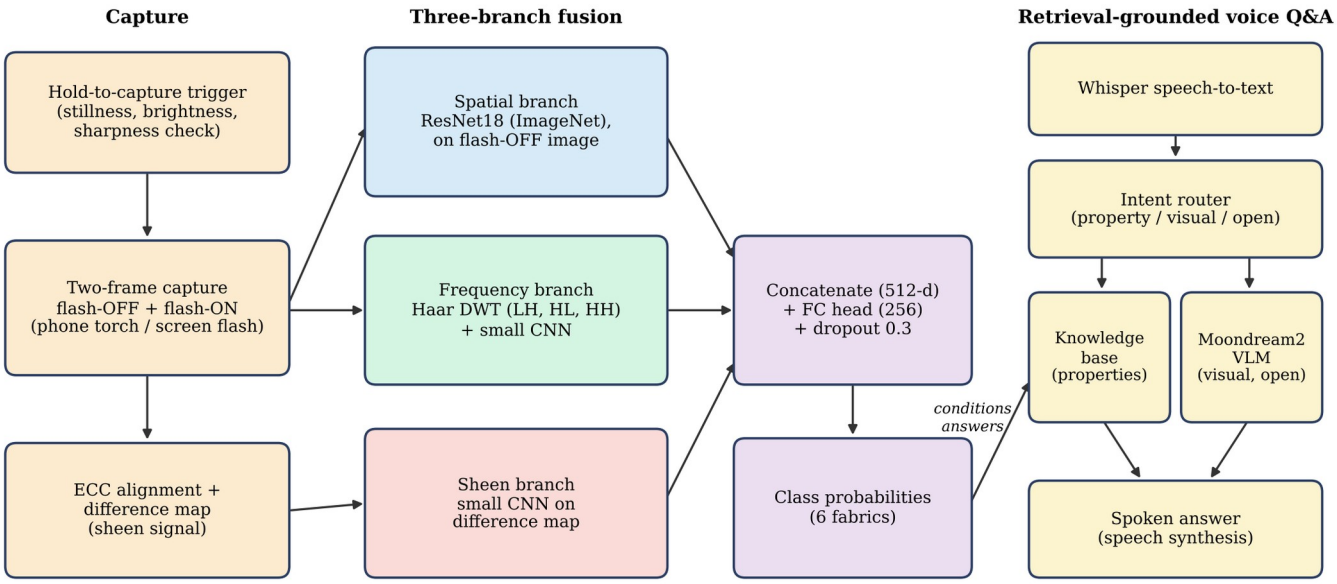


Figure 1: System architecture. Capture produces a flash pair and a sheen map; three branches are fused into class probabilities, which condition the knowledge-base and vision-language answers that are spoken back to the user.

3.4 Preprocessing and Sheen Map

Both frames are resized to 224×224. The brighter frame is registered to the darker one by ECC alignment under a Euclidean motion model, falling back to the unaligned frame if registration does not converge, and the absolute difference of the two gives the sheen map. The darker frame stands in for flash-OFF and the brighter for flash-ON, the same convention as in training. Images are normalised with ImageNet statistics. For training and evaluation on the iBUG dataset, which provides four illumination images per sample, the darkest and brightest of the four (by mean grey level) are used as the pair. This approximates the intended two-frame protocol with real, differently lit images, but it is a proxy (Section 6.2).

3.5 Three-Branch Classifier

The spatial branch is a ResNet18 [8] pretrained on ImageNet, chosen for its low computational cost, with a linear layer to 256 dimensions. The frequency branch applies a single-level Haar DWT to each colour channel of the flash-OFF image, stacks the nine detail sub-bands (LH, HL, HH per channel) and encodes them with a two-layer CNN to 128 dimensions; these sub-bands hold the fine repetitive structure of a weave that pooling in a standard CNN tends to smooth away [5, 4]. The sheen branch is a three-layer CNN on the difference map, also giving 128 dimensions. The three vectors are concatenated (512 dimensions) and passed through a 256-unit layer with ReLU and dropout of 0.3 to six output logits. A fixed Gabor filter bank (4 orientations × 3 wavelengths) is implemented as an alternative to the DWT branch, but has not been evaluated.

3.6 Training

Training runs for 20 epochs with batch size 16 and the Adam optimiser in two stages. The ResNet18 backbone is frozen for the first half while the new branches and head train at a learning rate of 1e-4, then unfrozen and fine-tuned at 1e-5. The best checkpoint by validation accuracy is kept. Because of the class imbalance, a class-weighted cross-entropy loss is used, with weight $N/(K \cdot n_c)$ for class c, giving about 0.32 for cotton, 1.16 for denim, 3.77 for silk, 2.10 for wool, 0.83 for polyester and 11.8 for leather. Training was done on a laptop CPU.

3.7 Hybrid Conversational Layer

Speech input uses Whisper [13] (small model, English). To limit hallucinated transcripts on background noise, a transcript is discarded when its mean no-speech probability exceeds 0.5 or it is implausibly long (over 25 words). A keyword-based intent router detects fabric-property intents: best season, summer, winter, durability, care, breathability, authenticity, what-is-this, uses, suitable ages, skin comfort, stretch and wrinkling, water resistance, cost, environmental impact, country of origin and blends (the extra intents are matched with word-boundary patterns to avoid accidental matches). A question about a blend is answered from the two highest class probabilities and notes that the classifier names a single fabric, so the care label gives the exact mix. For these the answer is retrieved from a hand-written knowledge base entry for the class with the highest probability, which makes it consistent and instant. A question that names another fabric (for example “does this look like real leather?” on a denim sample) is answered from the probabilities, for instance by stating that the classifier believes the sample is denim at 99 percent. Whether leather is real is deliberately not asserted from a photo. Questions about colour, pattern or texture go to Moondream2 [10] with the image alone, and any remaining question goes to Moondream2 with the retrieved facts and the class probabilities as context. The model output is cut to two sentences and checked for prompt echo, and when the top probability is below 70 percent an uncertainty note is appended. Multi-turn context passes the last two questions, but not earlier answers, so that the small model cannot copy itself; knowledge-base answers are stateless.

3.8 Web Implementation

The interface is a Flask server and a single-page browser client. The browser handles the camera (getUserMedia), the hold-steady trigger, the flash, microphone recording and speech synthesis for spoken answers; the server loads the classifier and Whisper once, loads Moondream2 in a background thread so the page is usable immediately, and exposes scan, transcribe and ask endpoints. Recording uses an energy-based voice activity detector: it measures the noise floor for 0.4 s, treats speech as an energy above three times that floor (and at least 0.02), and stops after 1 s of silence. Keyboard shortcuts (S, M, R) trigger scanning, listening and repeating the last answer, and every status change is spoken or announced to a screen reader. After each scan the server encodes the image for the vision-language model in a background thread, under its own lock so that scanning and speech recognition are never blocked. A zoom setting and a steadiness-sensitivity setting are exposed to the user. A dataset-sample mode runs the same pipeline on a stored sample for reliable demonstration. A terminal version of the same pipeline was used for the measurements in Section 5.

4. EXPERIMENTAL SETUP

4.1 Dataset

We use the iBUG Fabrics Dataset from Imperial College London [9]. It contains about 2,000 fabric and garment samples; for each sample a small surface patch was captured under four illumination conditions with a portable photometric-stereo sensor, in real clothes shops, and each sample comes with a metadata tag. Because it reflects fabrics found in shops, it is strongly imbalanced. Only six material folders were used: Cotton, Denim, Silk, Wool, Polyester and Leather. All other categories were excluded. This leaves 1,132 samples, all with the four illumination images (Table 1).

Table 1: Class distribution of the six-class subset and the held-out test split.

Class	Total	Train / val share	Test
Cotton	588	70 / 15 %	83
Denim	162	70 / 15 %	26
Silk	50	70 / 15 %	14
Wool	90	70 / 15 %	15
Polyester	226	70 / 15 %	26
Leather	16	70 / 15 %	5
Total	1132	794 / 169	169

4.2 Protocol and Metrics

Samples were split 70/15/15 with a fixed random seed into 794 training, 169 validation and 169 test samples, and the evaluation script regenerates the same split so that results are computed only on samples never seen in training. We report accuracy, per-class precision, recall and F1, and macro-F1. An earlier evaluation that scored 94% was run on training data and has been discarded; all figures below are held-out. With 169 test samples, the 95% margin of error of an accuracy near 80% is about six percentage points, and the leather test set has only five samples, so small differences must not be over-interpreted.

4.3 Ablation and Latency Protocol

The ablation trains four variants (spatial only, spatial + DWT, spatial + sheen, and the full model) with the same split, seed, class-weighted loss and schedule, changing only which branches are present. Latency was measured on a laptop CPU with no GPU, with the terminal version of the pipeline, over a small number of spoken questions.

5. RESULTS

5.1 Classification Performance

Overall accuracy of the full model is 81.7% (138 of 169) with a macro-F1 of 0.76 (Table 2). Two independent training runs of the weighted model both gave 82% accuracy. Denim, cotton and polyester are recognised well. Silk has perfect precision but a recall of only 0.64, with five of its fourteen test samples predicted as polyester (Figure 2), which is the visually and physically similar confusion that the sheen branch was intended to resolve.

Table 2: Per-class results of the full, class-weighted model on the held-out test set.

Class	Precision	Recall	F1	Support
Cotton	0.88	0.83	0.86	83
Denim	0.89	0.96	0.93	26
Silk	1.00	0.64	0.78	14
Wool	0.71	0.67	0.69	15
Polyester	0.61	0.88	0.72	26
Leather	1.00	0.40	0.57	5
Accuracy			0.82	169
Macro avg.	0.85	0.73	0.76	169

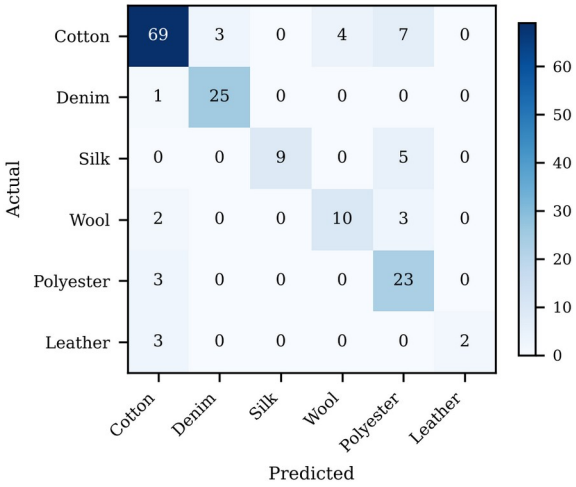


Figure 2: Confusion matrix on the held-out test set (169 samples).

Leather is the weakest class: only two of five test samples were recognised, and the other three were predicted as cotton. With 16 samples in total, leather is too rare for the model to learn a reliable signature.

5.2 Effect of Class Weighting

Class weighting was the most effective fix for the imbalance (Table 3): without it the model never predicted leather at all.

Table 3: Effect of class-weighted loss on the same test set.

Loss	Acc.	Macro F1	Leather R	Wool R
Unweighted	0.79	0.65	0.00	0.53
Class-weighted	0.82	0.76	0.40	0.67

5.3 Branch Ablation

Table 4 gives the ablation. Spatial-only reached 0.80 accuracy and 0.76 macro-F1; adding the DWT branch gave 0.78 and 0.73; adding the sheen branch gave 0.80 and 0.75; the full model gave 0.82 and 0.76. The full model is nominally best, but its margin over spatial-only is only a few samples out of 169, well inside the roughly six-point margin of error. The DWT branch on its own was nominally slightly worse than spatial-only. Silk recall was 0.64 for spatial-only and for the full model, and 0.71 with the

sheen branch alone (one more sample out of fourteen), while wool recall fell from 0.73 to 0.60 with the sheen branch. Leather recall was 0.40 in every variant. We therefore do not claim that either the frequency or the sheen branch improves accuracy.

Table 4: Branch ablation on the held-out test set (recall per class; Pol. = polyester, Lea. = leather).

Variant	Acc.	F1	Silk R	Wool R	Pol. R	Lea. R
Spatial only	0.80	0.76	0.64	0.73	0.85	0.40
+ DWT	0.78	0.73	0.64	0.73	0.85	0.40
+ Sheen	0.80	0.75	0.71	0.60	0.88	0.40
Full (both)	0.82	0.76	0.64	0.67	0.88	0.40

5.4 Behaviour of the Question-Answering Layer

The first design passed the image, the predicted class and the question directly to Moondream2. Testing on a denim sample exposed three failure modes. With instructions in the prompt, the model produced the same templated sentence about denim for every question. When asked to begin with yes or no it answered simply “Yes.” to all questions, including “How durable is this fabric?”. And without those instructions it contradicted itself, calling denim “a common summer fabric” in one turn and saying it can be worn in winter in the next. A small VLM has little reliable fabric knowledge and follows instructions poorly, so prompt tuning alone could not fix this, and we adopted the routed design of Section 3.7. Table 5 shows exchanges from a denim sample classified as denim with 99.3% probability. We did not measure the factual accuracy of the answers over a question set, and the knowledge base is hand-written, so these are qualitative examples only.

Table 5: Example spoken exchanges (denim sample).

Question (transcribed)	Route	Spoken answer
Can I wear this fabric during summer season?	KB	Not ideal. Denim is heavy and not very breathable, so it can feel hot in summer.
Can I wear this in winters?	KB	Yes, denim suits cooler weather, though layer up on very cold days as it is not strongly insulating.
Is this material durable?	KB	Denim is very durable and hard wearing.

5.5 Latency

Table 6 gives mean latencies on a laptop CPU. Capture, alignment and classification took 0.15 to 0.35 s per sample. Knowledge-base answers are essentially instant, while VLM answers dominate the delay. The samples are small and text-to-speech time grows with answer length, so the figures are indicative only. They come from the terminal version; latency of the browser version has not been measured.

Table 6: Mean per-question latency on CPU by answer route.

Route	STT	Answer	TTS	Total
KB (n = 3)	2.5 s	0.0 s	4.8 s	about 7 s
VLM (n = 2)	2.7 s	18.4 s	8.2 s	about 29 s

5.6 Live Capture Observations

The browser version was run end to end on the laptop with the real classifier, Whisper and the knowledge base. A scan produced class probabilities, the three images and the flash effect, and a spoken question (“In which weather can I wear this”) was answered from the knowledge base with the uncertainty note appended, because the top probability (polyester, 66%) was below 70%. The microphone worked once the browser's permission was granted; a dismissed permission prompt disables voice input, so the interface now explains how to re-enable it. Two observations are informal and are not measurements of accuracy. First, the original steadiness test, which compared raw frames against a limit of 3 grey levels, never triggered on the webcam, because sensor noise and hand tremor exceeded the limit. We reproduced this with a simulated noisy, shaking camera, and it led to the block-averaged measure, the sensitivity setting and the 8-second fallback of Section 3.2. Second, the predicted class changed between two captures (polyester at 66%, then cotton) after the fabric was held closer. The true fabric was not recorded, so these captures are not counted as errors, but the variation is consistent with the scale and lighting gap between webcam images and the close-up training patches (Section 6.3), and it motivated the zoom crop, which has not been evaluated. Figure 3 shows a later live scan of a woven fabric: the classifier gave cotton 59.9% and denim 26.2%, the uncertainty note was appended to the answer, and the weave is clearly visible in the zoomed crop. The true fabric was not recorded. The 19.5 s scan time in that screenshot came from a background image-encoding step that held a lock, which has since been fixed; classification itself normally takes 0.15 to 0.45 s.

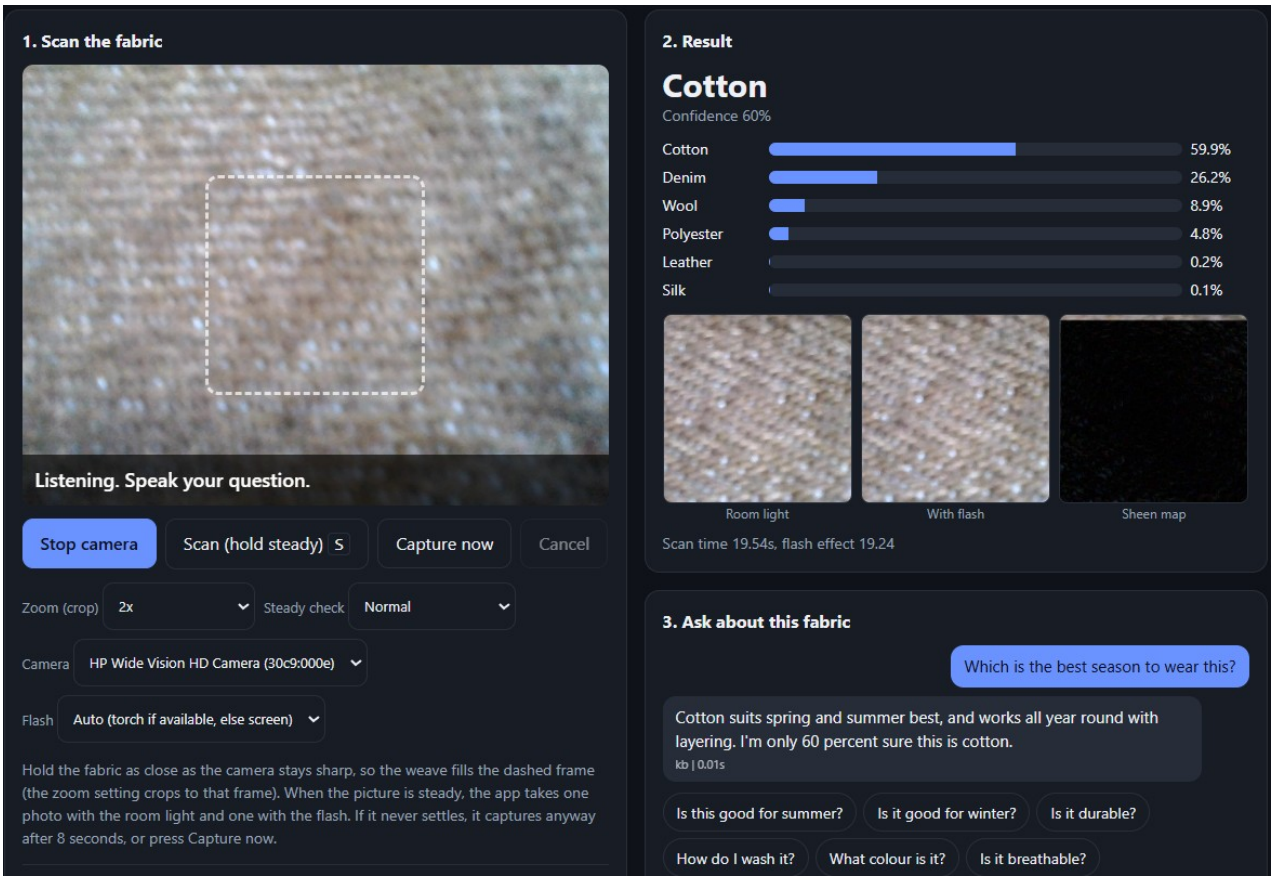


Figure 3: The web interface during a live scan on the laptop webcam (1280×720, screen flash, 2x zoom). Left: the live view with the fabric in the capture frame. Right: class probabilities, the room-light, flash and sheen images, and a question answered from the knowledge base. The true fabric was not recorded.

6. DISCUSSION

6.1 Key Findings

A pretrained ResNet18 fused with two small branches classifies six fabrics at 81.7% on held-out data, but the ablation does not support the claim that the DWT or sheen branch is responsible for that figure. The strongest measured effect in this study was not an architectural one: class-weighted training raised macro-F1 from 0.65 to 0.76 by making the model predict the rare classes. The other practical finding is about the conversational layer: routing property questions to curated facts removed the inconsistency that a small VLM showed on its own.

6.2 Why the Extra Branches May Not Help

The following are hypotheses that this study did not test. The ImageNet-pretrained ResNet18 may already capture most of the texture information in these images. Resizing to 224×224 may discard the high-frequency detail the DWT branch relies on. The dataset is small and imbalanced, so extra branches add parameters that cannot be learned reliably. And the dataset's differently lit images are only a proxy for a true flash-ON and flash-OFF pair, so the sheen signal may be weaker than in a real capture. Testing these needs a larger test set, repeated runs over several seeds, higher-resolution input and real flash captures.

6.3 Limitations

(1) The test set has 169 samples, so differences of a few points are within noise; leather (16 samples) and silk (50) are too rare for reliable per-class estimates.

(2) The classifier was trained on close-up sensor images with controlled lighting. Accuracy on webcam or phone captures of real garments has not been measured and must not be assumed to match the held-out figure; informal webcam scans gave low confidence and inconsistent classes (Section 5.6).

(3) The capture thresholds, the voice activity detector and the 60% and 70% confidence thresholds were set by hand and not validated.

(4) The knowledge base is small, hand-written and covers six classes, and the keyword router sends unusual phrasings to the weaker VLM route. The browser interface has been run end to end on one laptop with its built-in webcam, but not on other cameras or a phone, and not systematically.

(5) Latency was measured on CPU only and a VLM answer takes about half a minute. No usability study with visually impaired users has been carried out, and the system is a web prototype, not a native phone application.

7. CONCLUSION AND FUTURE WORK

We built and evaluated an end-to-end prototype that identifies a fabric from a flash pair and answers spoken questions about it. The fused classifier reaches 81.7% accuracy on a held-out set of six fabric classes, but the ablation shows no measurable benefit from the DWT or sheen branches on this data, so those contributions remain unproven rather than demonstrated. The

retrieval-grounded conversational layer gives consistent property answers where a small VLM alone did not.

Next steps are to compare a Gabor filter bank and ViT patch embeddings with the current choices; to repeat the ablations over several seeds and with higher-resolution frequency input; to collect more leather and silk samples; to measure accuracy and the domain gap on real webcam and phone flash captures; to evaluate a stronger VLM such as Qwen2-VL and expand the knowledge base; and to run a usability study with visually impaired participants.

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